

Multiple Systems

$$|\phi^+\rangle \langle \phi^+| =$$



$$|\psi^+\rangle \langle \psi^+| =$$



$$|\phi^-\rangle \langle \phi^-| =$$



$$|\psi^-\rangle \langle \psi^-| =$$



Product States

State $x \sim p$

State $y \sim \sigma$

$$(x, y) \sim p \otimes \sigma$$

Correlated and Entangled States

→ Correlated Classical States

$$\text{Ex.} \rightarrow \rho = \frac{1}{2} |0\rangle\langle 0| \otimes |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1| \otimes |1\rangle\langle 1|$$

→ Ensembles of Quantum States

$$\sum_{k=0}^{n-1} p_k |k\rangle\langle k| \otimes \rho_k$$

$$\rho_A = \sum_{b \in \Sigma_B} \frac{\|\phi_b\rangle\|^2}{\|\phi_b\rangle\|^4} \cdot |\phi_b\rangle\langle\phi_b| = \sum_{b \in \Sigma_B} |\phi_b\rangle\langle\phi_b|$$

$$= \rho_A = \sum_{b \in \Sigma_B} (I_A \otimes \langle b|) \rho (I_A \otimes |b\rangle)$$

$$\rho_B = \sum_{a \in \Sigma_A} (\langle a| \otimes I_B) \rho (|a\rangle \otimes I_B)$$

$$(A, B, C) \rightarrow \rho$$

$$\rho_A =$$

$$\rho_B =$$

$$\rho_C =$$

$$\rho_{AB} =$$

$$\rho_{BC} =$$

$$\rho_{AC} =$$

